

Harry Winston Sullivan

Curriculum Vitae

Department of Chemical Engineering
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Professional Summary Statement

- Chemical Engineering PhD student with a focus on computational science and engineering. My diverse background contains professional research in computer science, physics, chemical engineering, machine learning, and finance.
- Highly adept and creative researcher with a proven track record of swiftly conceptualizing, designing, and implementing innovative solutions to intricate engineering challenges. Possesses strong communication skills to effectively convey complex ideas.

Education

- 2024–Present **Ph.D. Chemical Engineering**, *University of Minnesota*, Minneapolis, MN, Advisors: **Prof. Sapna Sarupria** ([link](#)); **Prof. Ellad B. Tadmor** ([link](#)).
Generative modeling for materials design, including molecular crystal packing and advanced sampling methods for rare events in molecular simulations; diffusion models and geometric deep learning; statistical mechanics; non-equilibrium thermodynamics.
- 2023–2024 **M.S. Chemical Engineering**, *University of Utah*, Salt Lake City, UT, GPA: 4.0 (incomplete).
Neutron scattering analysis; molecular simulation; classical and quantum statistical mechanics; thermodynamics; Bayesian statistics and machine learning; continuum mechanics; catalysis; density functional theory.
- 2019–2023 **B.S. Computational Physics**, *University of Utah*, Salt Lake City, UT, GPA: 3.868.
Classical mechanics; electricity and magnetism; statistical mechanics; quantum mechanics; solid state physics; scientific computing.
- 2019–2023 **Minor, Computer Science**, *University of Utah*, Salt Lake City, UT.
Bayesian machine learning; uncertainty quantification; model validation; data structures and algorithms; neural networks.

Research Experience

- 2023–2024 **Graduate Research Assistant — Neutron Scattering, Quantum Molecular Simulation, and Machine Learning**, *University of Utah*, Salt Lake City, UT, **Dr. Michael Hoepfner**, Associate Professor ([link](#)).
Applied quantum statistical mechanics with variational optimization to infer interatomic forces from soft-matter microstructure.
- 2023 **Contract Researcher — Machine Learning for Cryptocurrency Markets**, *Blank Page Studios*, Remote, ([link](#)).
Three-month contract applying machine learning, scientific computing, and data analysis to cryptocurrency trading and NFT sales analytics; managed by Kenny Abitbol and Marcus Corbett.
- 2021–2023 **Undergraduate Researcher — Neutron Scattering, Molecular Simulation, and Machine Learning**, *University of Utah*, Salt Lake City, UT, **Dr. Michael Hoepfner**, Associate Professor ([link](#)).
Used classical statistical mechanics and Bayesian machine learning to infer interatomic forces from soft-matter microstructure.
- Summer 2021 **Research Assistant — Gravitational-Wave Data Analysis (LIGO)**, *University of Utah*, Salt Lake City, UT, **Dr. Yue Zhao**, Assistant Professor ([link](#)).
Refactored LIGO analysis workflows in Python and Fortran and authored standard operating procedures for future analyses.

Publications

- Feb 2026 **C. Zeng, H.W. Sullivan, T. Egg, M. M. Martirosyan, P. Höllmer, J. Jin, R. G. Hennig, A. Roitberg, S. Martiniani, E. B. Tadmor, M. Liu**, *MolCrystalFlow: Molecular Crystal Structure Prediction via Flow Matching*, ArXiv Preprint, ([link](#)).
- Oct 2025 **H.W. Sullivan, B.L. Shanks, M. Cervenka, M.P. Hoepfner**, *Physics-Informed Gaussian Process Inference of Liquid Structure from Scattering Data*, *J. Phys. Chem. B* 2025, 129(45), 11802–11815, ([link](#)).
- Apr 2025 **B.L. Shanks*, H.W. Sullivan, P.J. Jungwirth, M.P. Hoepfner**, *Experimental Evidence of Quantum Drude Oscillator Behavior in Liquids Revealed with Machine Learning Assisted Iterative Boltzmann Inversion*, *J. Chem. Phys.* 162, 164501 (2025), ([link](#)).
- Dec 2024 **B.L. Shanks, H.W. Sullivan, M.P. Hoepfner**, *Bayesian Analysis Reveals the Key to Extracting Pair Potentials from Neutron Scattering Data*, *J. Phys. Chem. Lett.* 2024, 15, 12608–12618, ([link](#)).
- Mar 2024 **B.L. Shanks, H.W. Sullivan, A.R. Shazed, M.P. Hoepfner**, *Accelerated Bayesian Inference for Molecular Simulations using Local Gaussian Process Surrogate Models*, *J. Chem. Theory Comput.* 2024, 20, 3798–3808, ([link](#)).

Contributed Talks and Posters

- Mar 2025 **Generative Modeling of Boltzmann Distributions Using Denoising Diffusion Models**, *Institute of Organic Chemistry and Biochemistry (IOCB Prague)*, *P. Jungwirth Group*, Prague, Czech Republic.
- Jul 2024 **Estimation and Uncertainty Quantification of Experimental Radial Distribution Functions Using Non-Stationary Gaussian Processes**, *Foundations of Molecular Modeling and Simulation*, *U.S. Department of Energy (DOE)*, Salt Lake City, UT.
Poster presentation.
- Jan 2024 **Current Challenges in Reconstructing Classical Force Fields from Neutron Scattering Data**, *DOE EFRC: Multi-scale Fluid–Solid Interactions in Architected and Natural Materials*, Salt Lake City, UT.
- Nov 2023 **Bayesian Inverse Quantum Theory**, *Theoretical Chemistry Department*, *University of Utah*, Salt Lake City, UT.
- Sep 2023 **Learning Interatomic Forces from Fluid Structure with Machine-Learning-Accelerated Bayesian Optimization**, *American Chemical Society (ACS)*, Laramie, WY.
- Mar 2022 **Bayesian optimized force fields enabled by a radial distribution function surrogate model**, *Recent Advances in Machine-Learning-Accelerated Molecular Dynamics*, *Centre Européen de Calcul Atomique et Moléculaire (CECAM)*, Trieste, Italy.
Poster presentation. Presented with Brennon L. Shanks.